Project Design Phase
Problem – Solution Fit Template

Date	15 February 2025
Team ID	
Project Name	**Credit Card Approval Prediction**
Maximum Marks	2 Marks

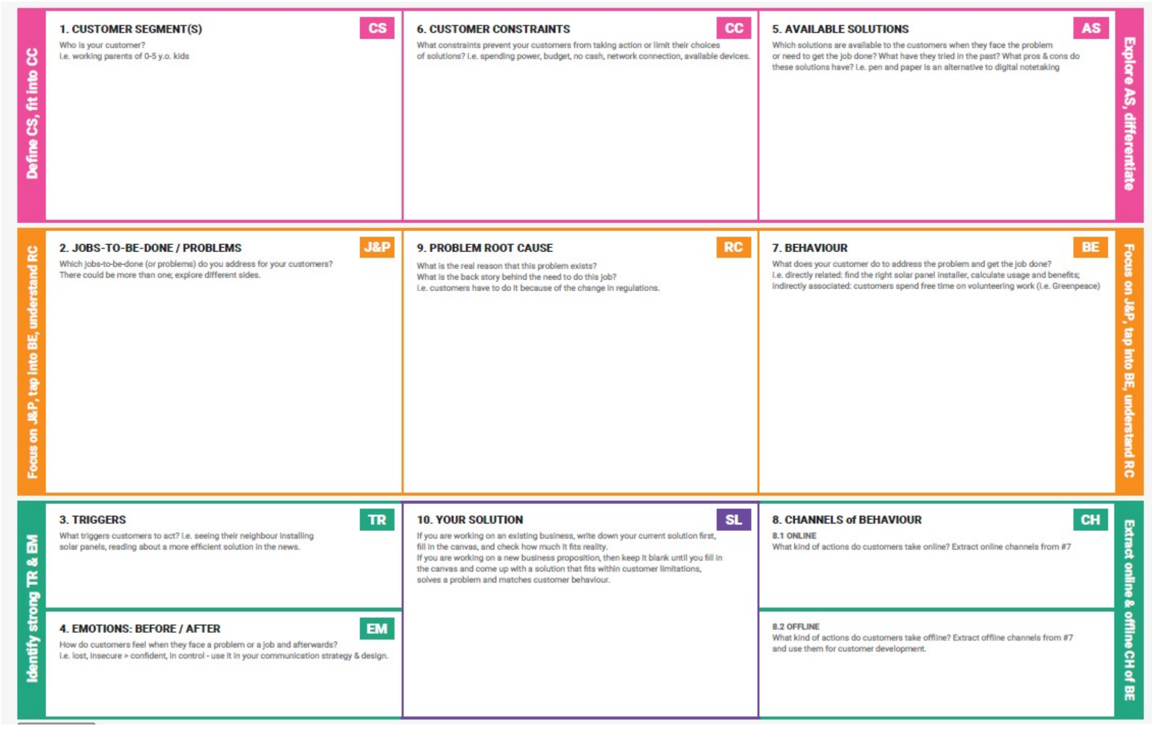
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



Filled Canvas — Credit Card Approval Prediction

1. CUSTOMER SEGMENT(S) *Who is your customer?* • Retail & digital banks and fintech card issuers • Credit risk / underwriting teams within those institutions • Loan officers and credit analysts who review applications • Compliance & fair-lending officers who audit decisions	**2. JOBS-TO-BE-DONE / PROBLEMS** *Which jobs or problems do you address?* • Decide quickly whether to approve or decline a credit card application • Minimize default and credit-loss exposure from bad approvals • Reduce manual review time and inconsistent human judgment • Stay compliant with fair-lending regulations while scaling volume	**3. TRIGGERS** *What triggers customers to act?* • A new application arrives and needs a same-day decision • Rising default rates push leadership to demand better risk models • Regulators or auditors flag inconsistent manual decisions • Competitors launch instant-approval card products
6. CUSTOMER CONSTRAINTS *What limits their choice of solutions?* • Must comply with fair-lending / anti-discrimination law (e.g. ECOA) • Need explainable, auditable decisions, not a pure black box • Limited budget and IT capacity to integrate new scoring systems • Data privacy rules restrict which applicant data can be used	**9. PROBLEM ROOT CAUSE** *What is the real reason this problem exists?* • Traditional bureau scorecards are static and slow to reflect new risk patterns • Manual underwriting rules don't scale and vary reviewer to reviewer • Result: too many risky approvals (losses) or too many good customers declined (lost revenue)	**5. AVAILABLE SOLUTIONS** *What have they tried? Pros & cons?* • FICO / bureau credit scores — reliable but generic, not tailored to issuer's portfolio • Manual underwriting rules — explainable but slow and inconsistent • Legacy logistic-regression scorecards — auditable but low accuracy • Third-party approval APIs — fast but costly and low transparency
7. BEHAVIOUR *What do customers do to get the job done?* • Pull the applicant's credit bureau report and internal transaction history • Apply rule-based checks (income, existing debt, utilization) before escalation • Escalate borderline applications to a senior underwriter for manual review	**10. YOUR SOLUTION** *Current or proposed solution* • An ML model that scores each application on approval / default likelihood using applicant and bureau data • Automates clear-cut approvals and declines instantly, routes only borderline cases to human review • Includes an explainability layer so	**4. EMOTIONS: BEFORE / AFTER** *How do they feel before and after?* • Before: anxious about hidden risk, overwhelmed by backlog, unsure decisions are consistent • After: confident in a defensible, data-driven decision; in control of portfolio risk and turnaround

	decisions can be audited and defended to regulators	
8.1. CHANNELS — ONLINE *Online actions taken* • Internal loan/credit processing system used to submit and track applications • Bureau data APIs (Experian, Equifax, TransUnion) queried for credit history • Internal risk dashboards where analysts review model scores	**8.2. CHANNELS — OFFLINE** *Offline actions taken* • In-branch application intake and document verification • Phone follow-ups with applicants for missing information • Underwriting committee meetings for high-value or disputed cases	

Purpose / Vision: Use predictive modeling to make credit card approval decisions faster, fairer, and more accurate — reducing losses while approving more creditworthy customers.

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe